Final Progress Report

Tanay Mihani, Keith Fernandes, Ayush Mahapatra

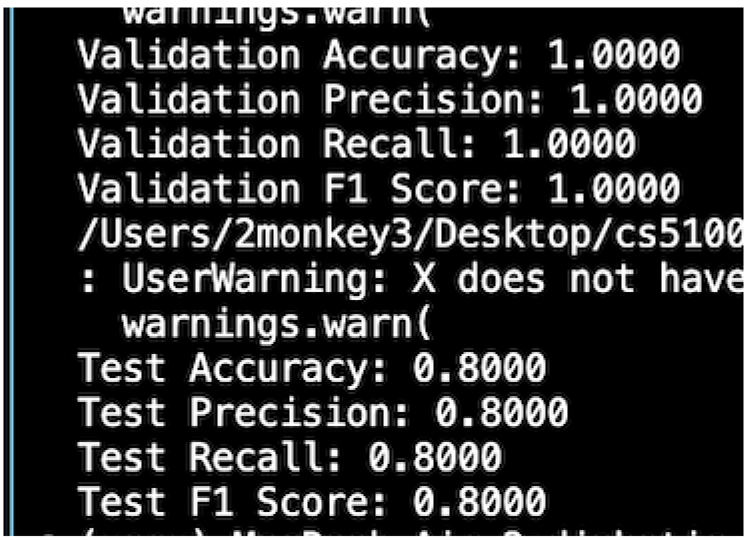
Here are the steps that we have accomplished:

1. Image preprocessing - Keith Fernandes
2. GLCM Features Extraction and Data Splitting - Tanay Mihani
3. Full, initial Feature Selection and LightGBM classifier with GA optimization (Blended crossover) with evaluations of the model - Ayush Mahapatra

Here are the things we need to put in:

1. A baseline model with no feature selection or hyperparameter tuning with GA algorithm to compare the impact of the GA algorithm
2. Have a complete README.md file for users to run the code properly
3. Find out the feature reduction involved as well

This is our current results:



From the results above on the validation and testing set results, you can see that our model is overfitting. We will need to refer to the paper to make sure our feature selection and hyperparameter tuning is being done properly. This could also be the result of the fact that we might not be able to run as many epochs and have as many populations for our code to run on a normal laptop. This could go into future implications to keep in mind.

Overall, we are in a good position. I believe we can accomplish these things before the deadline of the project.

Here is our Github: https://github.com/ayumahapat0/diabetic_retinopathy_prediction/tree/Ayush/GA

Here are the commands to run the code:

1. Python3 IMAGE_PREPROCESSING.py
2. Python3 splits/data-splitting.py
3. Python3 GLCM_Feature_extraction.py
4. Python3 GA_Optimization.py